



SIMULATION OF VEHICLE IN UNREAL GAME ENGINE USING EQUATIONS OF MOTION, RUNGE-KUTTA AND LINE TRACING METHODS TO SOLVE MOTION AND SCAN UNREAL ENGINE TERRAIN UNDER WHEELS

SIMULÁCIA VOZIDLA V HERNOM PROSTREDÍ UNREAL ENGINE
S POUŽITÍM POHYBOVÝCH ROVNÍC, RUNGE-KUTTA A LINE-TRACE
METÓD PRE RIEŠENIE POHYBU A SKENOVANIE TERÉNU
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Abstract: The paper is aimed on description and results of simulation of a vehicle in Unreal Engine environment. We created a quite simple vehicle model we plan to use in subsequent simulations and optimizations by genetic algorithm method. The main idea was to create the model as simple as possible, to decrease computer performance needs as much as possible, but on the other hand, we required a visually real motion of the vehicle model on a terrain in Unreal Engine. The model is defined by equations of motion implemented in C++ and its dynamical behavior can be verified on a computer's monitor or in virtual reality using HTC Vive headset.

Key words: vehicle simulation, dynamical analysis, Runge-Kutta, Unreal Engine, line tracing, equations of motion, C++, virtual reality

Abstrakt: Článok je orientovaný na popis a výsledky simulácie vozidla v prostredí Unreal Engine. Vytvorili sme pomerne jednoduchý model vozidla, ktorý plánujeme následne použiť na simuláciu a optimalizáciu genetickým algoritmom. Základná myšlienka bola vytvoriť model tak jednoduchý ako je to len možné, ale na druhej strane sme požadovali vizuálne reálny pohyb modelu vozidla na teréne v Unreal Engine systéme. Model je definovaný pohybovými rovnicami, implementovanými v jazyku C++, pričom jeho dynamické správanie môže byť verifikované na monitore počítača alebo vo virtuálnej realite s použitím HTC Vive náhlavnej súpravy.

Kľúčové slová: simulácia vozidla, dynamická analýza, Runge-Kutta, Unreal Engine, line tracing, pohybové rovnice, C++, virtuálna realita

1 INTRODUCTION

The Unreal Engine [1] is an environment historically focused on game development. One of the most known game, created using this environment, is Unreal Tournament – one of the first 3D first person shooting game. In present the environment is freely available under a very friendly license, and it contains a development environment connected with Microsoft Visual Studio. We used Visual Studio to write the motion equations and all the necessary code to display and control the vehicle's objects using a steering wheel and a keyboard, and

output the results. The final solution of the project will consist of a steel frame with a play seat, steering wheel, and a computer with the monitor and the headset.

The Unreal Engine (UE4) is a game engine, which from point of view of the paper means that it is focused on real-time side of the simulation. In other words, if a load of the system is higher than actual system possibilities, the system will automatically decrease resolution or FPS (frames per second) rate to achieve real-time simulation. This can be a problem if a numerical integration method is used to compute vehicle's dynamical parameters.

2 MATERIALS AND METHODS

A simulation environment we created consists of the terrain created by the means of Unreal Editor and imported objects from Creo Parametric 2.0. The vehicle consists of a chassis, wheels, back lights and glass cabin with sets. All the vehicle parts were created in Creo Parametric (Fig.1), exported as STEP and converted into FBX file format.

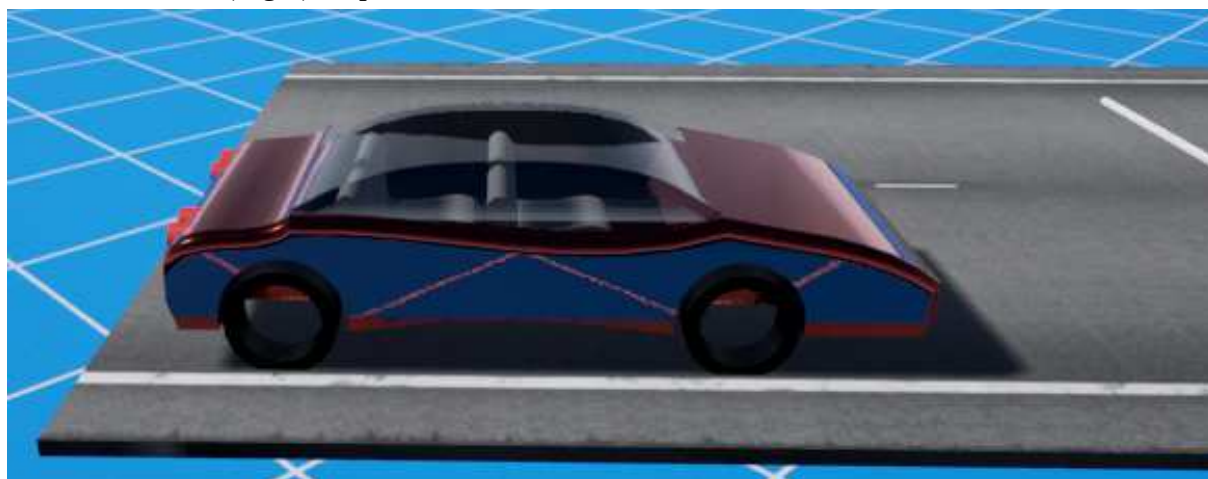


Figure 1 Vehicle model and road with cross obstacle. The look is a result of materials applied in Unreal Editor.

The Unreal Engine was used in order to visualize motion of the vehicle, and to create and use the terrain by a quite simply way. Motion of the vehicle is described with equations of motion, solved by Runge-Kutta 4th order method in Tick(double deltaTime) method. To get distances of the wheels under the terrain we used line tracing method available in UE4 API and forum [2]. These distances were used in the equations of motion of the vehicle. The main idea was to create the model as simple as possible, to decrease computer performance needs as much as possible, but on the other hand, we required an ability to visually evaluate a motion of the vehicle model. This is why we created a model that:

- allows motion of wheels in vertical direction only, considering the vehicle's frame,
- uses no tire and only one spring-dumper unit to simulate cushioning,
- supposes continuous contact of a wheel with the terrain (Fig.2), except the case when a spring's length should be higher than the free length of the spring,
- takes into the account only these forces (Fig.3): F_h – drive forces, F_b – brake forces, F_a – aerodynamic force, G – gravity force, D'Alembert forces, F_p – spring forces, F_y – side forces on the wheels,

- has all the forces dependent on forces in the springs, which are, in this case, the same as normal forces,
- has the wheels which allow no rotation and the drive forces are applied directly on the wheels, without using an engine characteristic to simplify planned simulations and optimizations.



Figure 2 Playback of drive on imported terrain



Figure 3 Selected forces acting on the vehicle

3 RESULTS AND DISCUSSION

The result of the above mentioned rules and requirements is an computer application written in C++, Visual Studio Community Edition and Unreal Editor, that has for the present following modes:

Free drive mode

The mode enables the user to test her/his model (= equations of motion). This mode with steering wheel and pedals is similar to a racing game, however the mode allows recording of a drive.

Playback mode

The mode allows playback of a drive recorded in free drive mode. The forces are visualized by arrows (Fig.4) and steering wheel paddle shifters allow changing the view angle in geographic coordinate system (to rotate around vehicle).

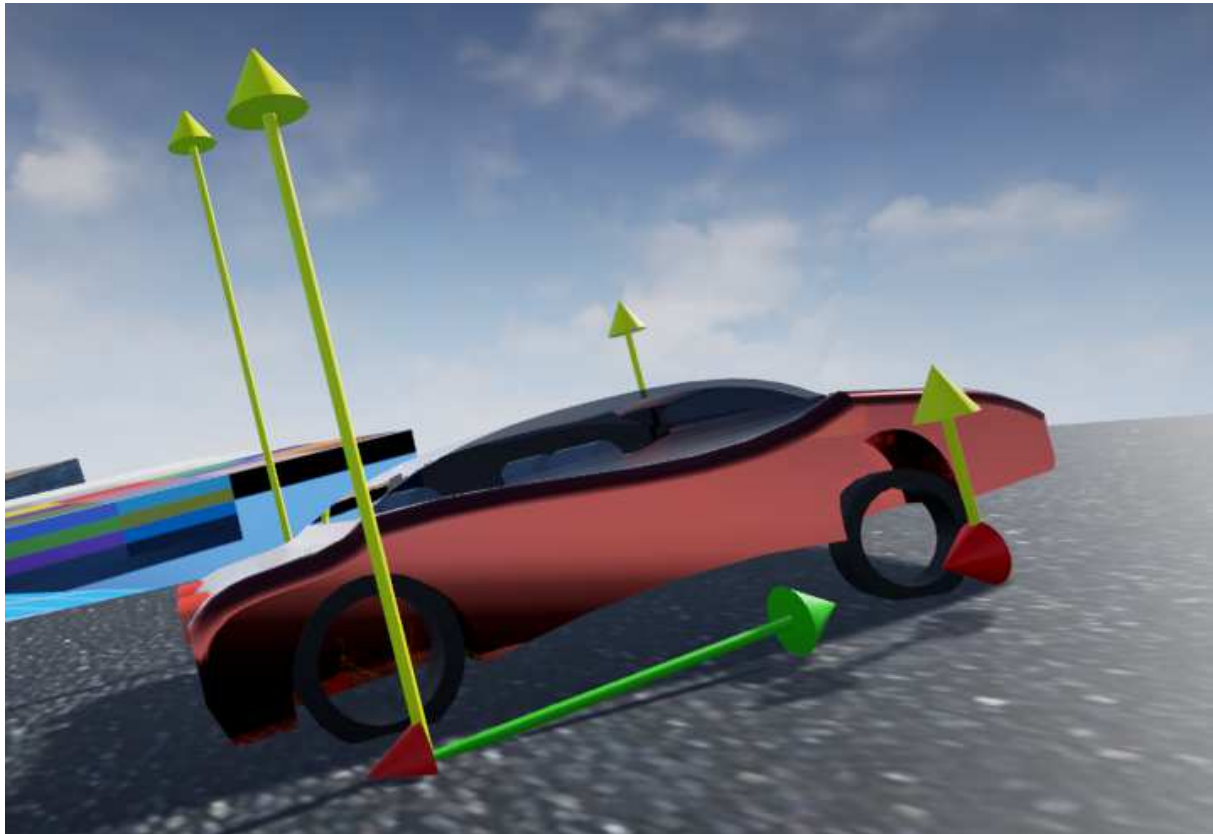


Figure 4 Visualized forces on a jump ramp in the playback mode

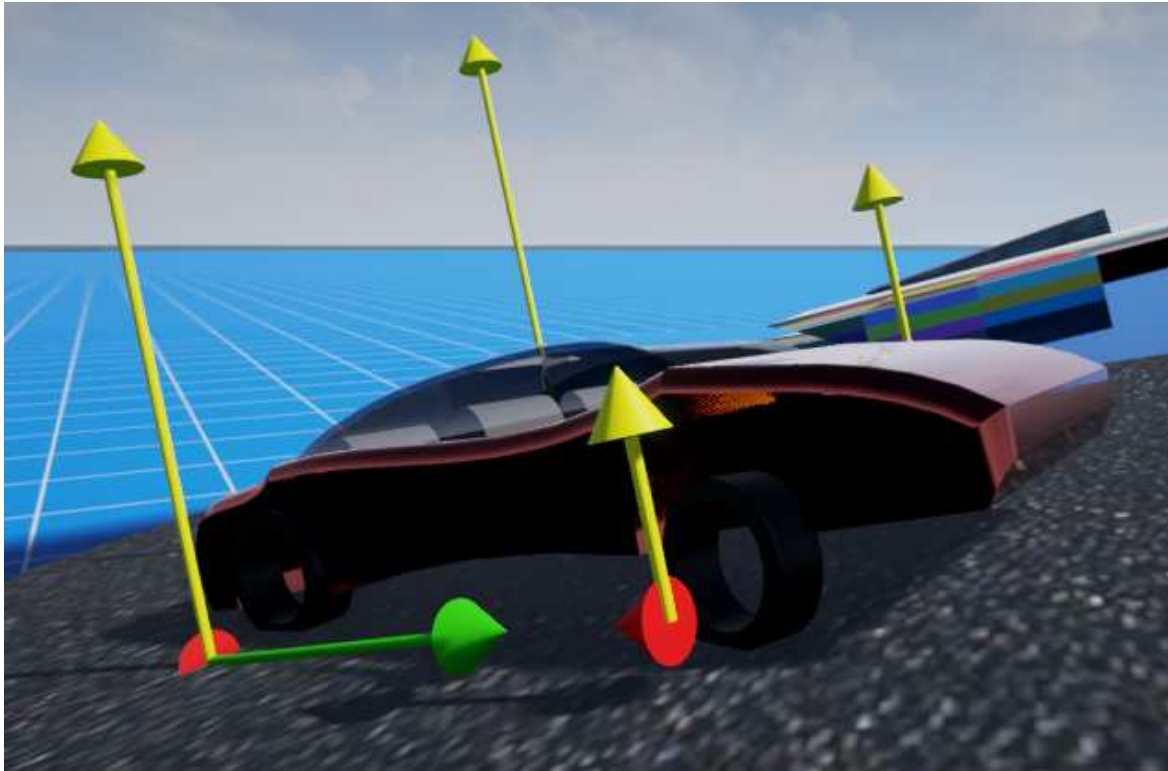


Figure 5 Visualized forces on a jump ramp in the playback mode

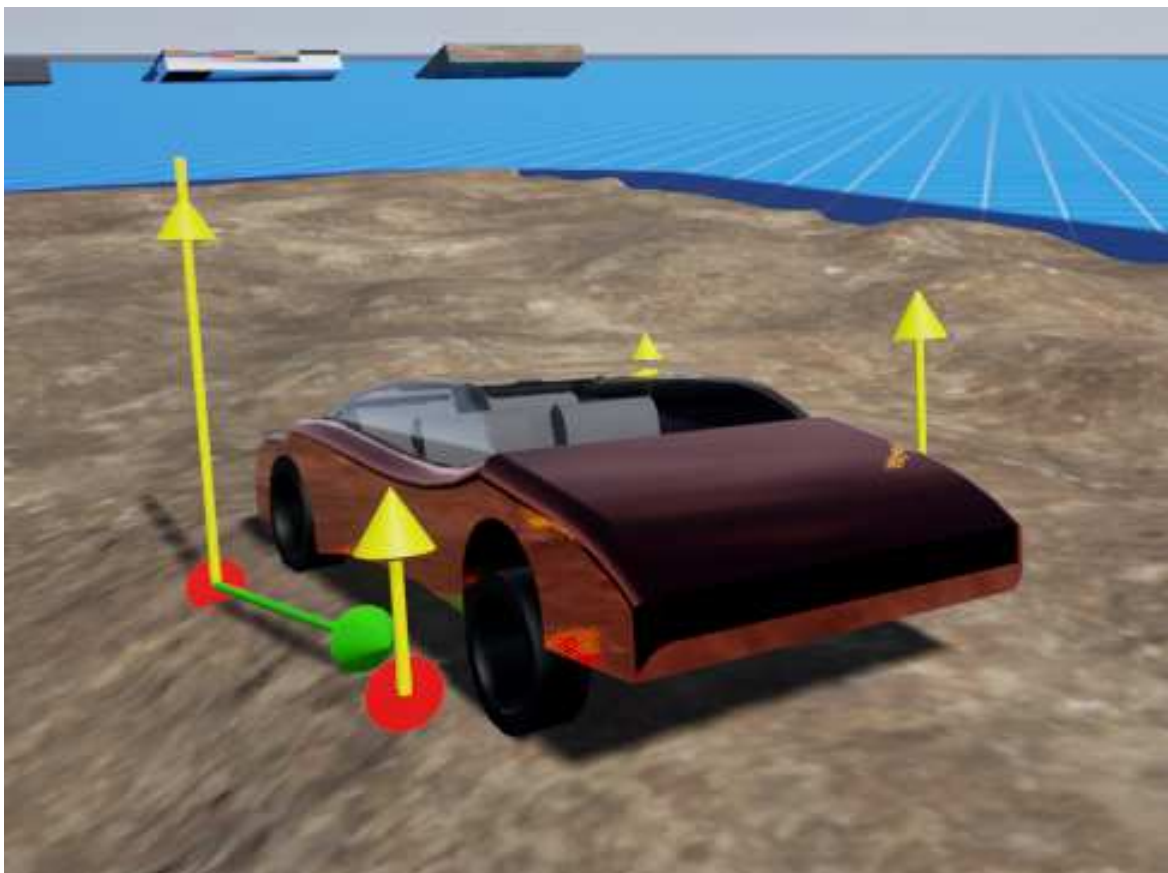


Figure 6 Visualized forces on an imported terrain in the playback mode

The simulation system in this configuration is fully usable, ready to improvements and prepared to be used as a base for prepared research of simulation and optimization. Thanks to the Unreal Engine the quality of the visualization is gorgeous and quite simply usable (see clouds on Figs. 2, 4, 5, metallic car paint and reflection on Figs. 2, 3, 6). If a more complex world is required it can be done by DIY way in Unreal Editor, or a huge number of object can be buy from third parties at Unreal Marketplace [3].

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